

Body size as a metric for the affordable world

Xinran Feng, Shan Xu, Yuannan Li, Jia Liu 

Department of Psychology & Tsinghua Laboratory of Brain and Intelligence, Tsinghua University, Beijing, China • Faculty of Psychology, Beijing Normal University, Beijing, China

 https://en.wikipedia.org/wiki/Open_access Copyright information

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2

February 19, 2024 (this version)

Reviewed preprint version 1

October 24, 2023

Sent for peer review

August 14, 2023

Posted to preprint server

May 6, 2023

Abstract

The physical body of an organism serves as a vital interface for interactions with its environment. Here we investigated the impact of human body size on the perception of action possibilities (affordances) offered by the environment. We found that the body size delineated a distinct boundary on affordances, dividing objects of continuous real-world sizes into two discrete categories with each affording distinct action sets. Additionally, the boundary shifted with imagined body sizes, suggesting a causal link between body size and affordance perception. Intriguingly, ChatGPT, a large language model lacking physical embodiment, exhibited a modest yet comparable affordance boundary at the scale of human body size, suggesting the boundary is not exclusively derived from organism-environment interactions. A subsequent fMRI experiment revealed evidence of affordance processing exclusively for objects within the body size range, but not for those beyond. This suggests that only objects capable of being manipulated are the objects capable of offering affordance in the eyes of an organism. In summary, our study suggests a novel definition of object-ness in an affordance-based context, advocating the concept of embodied cognition in understanding the emergence of intelligence constrained by an organism's physical attributes.

eLife assessment

This paper presents **valuable** findings that shed light on the mental organisation of knowledge about real-world objects. It provides diverse, if **incomplete** and tentative, evidence from behaviour, brain, and large language models that this knowledge is divided categorically between relatively small objects (closer to the relevant scale for direct manipulation) and larger objects (further from the typical scope of human affordances for action).

Introduction

Man is the measure of all things. - Protagoras

The assertion by the ancient Greek philosopher Protagoras highlights the notion that reality is defined by how the world is perceived by humans. A contemporary interpretation of this statement is the embodied theory of cognition (e.g., [Chemero, 2013](#); [Gallagher, 2017](#); [Gibbs,](#)

2005; Wilson, 2002 [↗](#); Varela et al., 2017 [↗](#)), which, diverging from the belief that size and shape are inherent object features (e.g., de Beeck et al., 2008 [↗](#); Konkle & Oliva, 2011 [↗](#)), posits that human body scale (e.g., size) constrains the perception of objects and the generation of motor responses. For instance, humans evaluate the climbability of steps based on their leg length (Mark, 1987 [↗](#); Warren, 1984 [↗](#)), and determine the navigability of apertures according to the critical aperture-to-shoulder-width ratio (Warren & Whang, 1987 [↗](#)). Additionally, grasping strategies have been shown to be contingent upon object size relative to one's body (Cesari & Newell, 2000 [↗](#); Newell et al., 1989 [↗](#)) or hand size (Castiello et al., 1993 [↗](#); Tucker & Ellis, 2004 [↗](#)). However, the question of how object perception is influenced by the relative size of objects in relation to the human body remains open. Specifically, it is unclear whether this relative size simply acts as a continuous variable for locomotion reference, or if it affects differentiating and organizing object representation based on their ensued affordances.

To underscore the latter point, Gibson (1979) [↗](#), the pioneer of embodied cognition research, stated that “Detached objects must be comparable in size to the animal under consideration if they are to afford behavior (p.124).” This implies that an object's affordance, encompassing all action possibilities offered to an animal, is determined by the object's size relative to the animal's size rather than its real-world size. For instance, in a naturalistic environment, such as a picnic scene shown in **Fig. 1a** [↗](#), there may exist a qualitative distinction between objects within (the objects with warm tints in **Fig. 1a** [↗](#)) and beyond (those with cold tints) the size range of humans. Only objects within the range, such as the apple, the umbrella, and the bottles, may afford actions, while those beyond this range, such as the trees and the tent, are largely viewed as part of the environment. Consequently, visual perception may be ecologically constrained, and the body may serve as a metric that facilitates meaningful engagement with the environment by differentiating objects that are accessible for interactions from those not. Further, grounded cognition theory (see Barsalou, 2008 [↗](#) for a review) suggests that the outputs of such differentiation might transcend sensorimotor processes and integrate into supramodal concepts and language. From this perspective, we proposed two hypotheses: first, the affordance of objects will exhibit a qualitative difference between objects within and beyond the size range of an organism's body; second, affordance-related neural activity will emerge exclusively for objects within the organism's size range.

To test these hypotheses, we first measured the affordance of a diverse array of objects varying in real-world sizes (e.g., Konkle, & Oliva, 2011). We found a dramatic decline in affordance similarity between objects within and beyond the human body size range, as these objects afforded distinct sets of action possibilities. Notably, the affordance boundary varied in response to the imagined body sizes and showed supramodality. It could also be attained solely through language, as evidenced by the large language model (LLM), ChatGPT (OpenAI, 2022 [↗](#)). A subsequent fMRI experiment corroborated the qualitative difference in affordances demarcated by the body size, as affordances of objects within humans' size range, but not those beyond, were represented in both dorsal and ventral visual streams of the brain. This study advances our understanding of the role of body size in shaping object representation and underscores the significance of body size as a metric for determining object affordances that facilitates meaningful engagement with the environment.

Results

To illustrate how human body size affects object affordances with different sizes, we first characterized the affordances of a set of daily objects. In each trial, we presented a matrix consisting of nine objects and asked participants to report which objects afforded a specific action (e.g., sit-able: a chair, a bed, a skateboard, but not a phone, a laptop, an umbrella, a kettle, a plate, or a hammer) (**Fig. 1b** [↗](#)). In this task, there were 14 actions commonly executed in daily life and 24 object images from the THINGS database (Hebart et al., 2019 [↗](#)), with sizes ranging from size

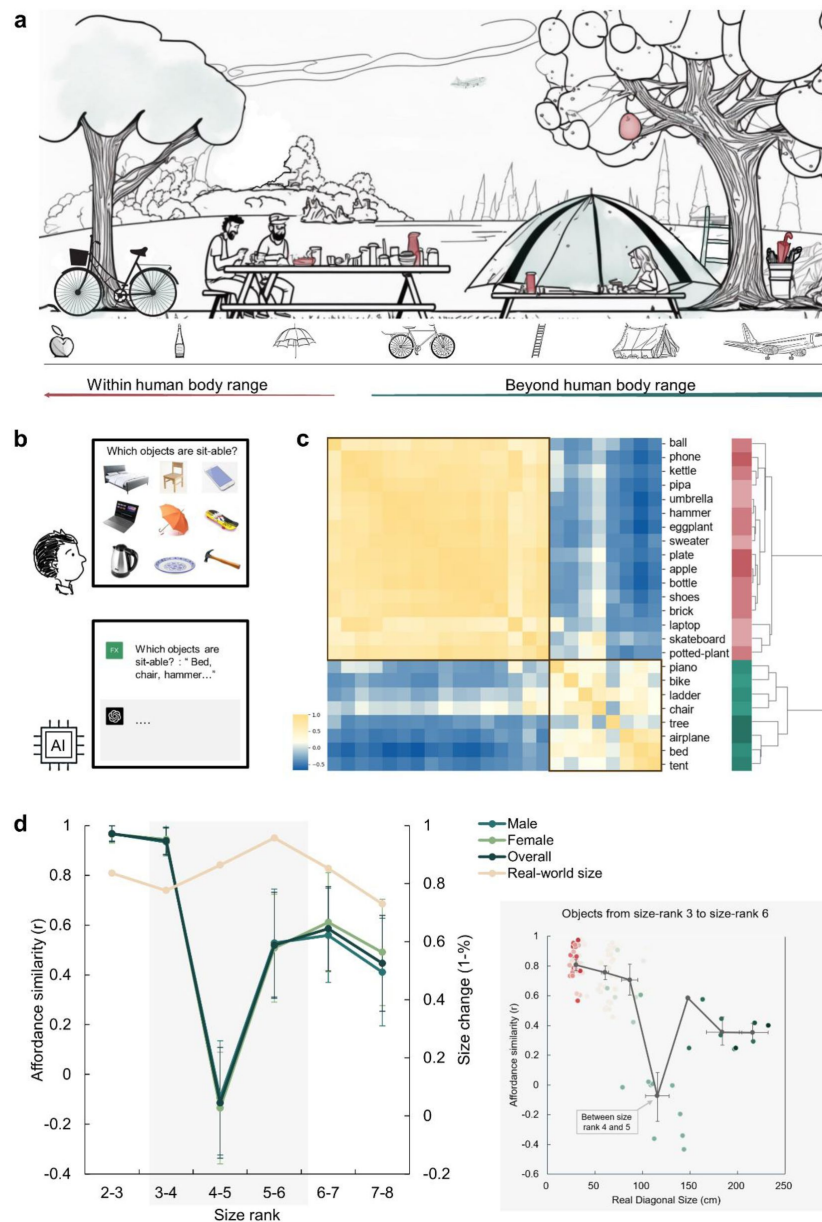


Fig 1.

An affordance boundary in the affordable world.

a, An illustration of a picnic scene, featuring objects of various sizes relative to human body. Example objects within the normal body size range are painted red, and those beyond green. We hypothesized qualitative differences between perceived affordances of these two kinds of objects. **b**, A demonstration of the object-action relation judgement task for human participants (top) and AI models (bottom). The question in the task for human participants was presented in Chinese. **c**, The representational similarity matrix (RSM) for objects based on human rating of affordance similarity. Object sizes are denoted with red to green. Two primary clusters emerged in the clustering analysis of the similarity pattern are outlined with black boxes. **d, Left panel:** The overall affordance similarity and that of each gender (left y-axis) as well as real-world size similarity (right y-axis) between neighboring size ranks. The error bars represent the standard error (SE). **d, Right panel:** The point clouds of pairwise correlations between objects from the same rank or neighboring ranks. Each colored dot represents the affordance similarity (y-axis) and the average real-world size (x-axis) of a specific object pair. The grey dots indicate the averaged size (x-axis) and pairwise similarity (y-axis) of object pairs in different rank compositions. Left to right: both from size rank 3, from size rank 3 and 4, both from size rank 4, from size rank 4 and 5, both from size rank 5, from size rank 5 and 6, and both from size rank 6. The horizontal error bars represent 95% confidence interval (CI) of the averaged object size in each pair, and the vertical error bars denote the CI of pairwise affordance similarity.

rank 2 to 8 according to Konkle and Oliva (2011) [\[1\]](#)'s classification. These objects covered real-world sizes from much smaller (17 cm on average, rank 2) to orders of magnitude larger (5,317 cm on average, rank 8) than the human body size (see Methods for details). Consequently, affordances for each object were indexed by a 14-dimensional action vector, with the value for each dimension representing the percentage of participants who agreed on a certain action being afforded by the object (e.g., 88% for the action of grasping on a hammer indicating 88% of participants agreed that a hammer affords grasping). Supplementary Fig. S1 shows the affordances of two example objects.

An affordance similarity matrix was then constructed where each cell corresponded to the similarity in affordances between a pair of objects (**Fig. 1c** [\[1\]](#)). A clustering analysis revealed a two-cluster structure. Visual inspection suggested that the upper-left cluster consisted of objects smaller than human body size (red labels), and the lower-right cluster contained objects larger than human body size (green labels). Critically, the between-cluster similarity in the affordance similarity matrix approached zero, suggesting a division in affordances located near the body size. To quantify this observation, we calculated the similarity in affordances between each neighboring size rank. Indeed, we identified a clear trough in affordance similarity, dropping to around zero, between size rank 4 (77cm on average) and 5 (146cm on average), which was significantly smaller than that between size rank 3 and 4 ($Z = 3.91, p < .001$) and that between size rank 5 and 6 ($Z = 1.66, p = .048$). This trough suggested an affordance boundary between size rank 4 and 5, while affordance similarities between neighboring ranks remained high ($r_s > 0.45$) and did not significantly differ from each other ($p_s > 0.05$, all $BF_{10} < 10$) on either side of the boundary (**Fig. 1d** [\[1\]](#), left panel, green lines). This pattern was evident for both genders, indicating no gender difference. Note that the abrupt change in affordance similarity across the boundary cannot be explained by changes in objects' real-world size, as the similarity in objects' real-world size was relatively stable across ranks, without any trend of a trough-shape curve (**Fig. 1d** [\[1\]](#), left panel, yellow line). Intriguingly, rank 4 and rank 5 correspond to 80 cm to 150 cm, a boundary situated between these two ranks is within the range of the body size of a typical human adult. This finding suggested that objects were classified into two categories based on their affordances, with the boundary aligning with human body size.

To better locate the boundary, we focused on the affordance similarity between individual objects within size rank 3 to 6 (approximately ranging from 30cm to 220cm in real-world size, the area with grey shade in **Fig. 1d** [\[1\]](#)), where the trough-shape curve was identified. Specifically, we traversed all pairs of objects with similar real-world diagonal sizes (from either the same rank or from neighboring ranks), calculated their average real-world size as an index of the approximate location of the boundary between this pair of objects, and plotted the affordance similarity against the average real-world size of each object pair. As shown in the inset (grey box) of **Fig. 1d** [\[1\]](#), consistent with the rank-wise analysis, the abrupt decrease in affordance similarity exclusively happened between objects from size rank 4 and 5 (light green dots). The averaged real-world size in these object pairs was 104 cm (95% CI, 105 to 130 cm) and the affordance similarity in such object pairs was around zero. This result further narrowed the location estimation of the boundary, and demonstrated that the affordance boundary persisted at the level of individual objects.

One may argue that the location of the affordance boundary coincidentally fell within the range of human body size, rather than being directly influenced by it. To rule out this possibility, we directly manipulated participants' body schema, referring to an experiential and dynamic functioning of the living body within its environment (Merleau-Ponty & Smith, 1962 [\[2\]](#)). This experimental approach was able to establish a causal link between body size and affordance boundary, as other potential factors remained constant. Specifically, we instructed a new group of participants to imagine themselves as small as a cat (typical diagonal size: 77cm, size rank 4, referred to as the "cat condition"), and another new group to envision themselves as large as an elephant (typical diagonal size: 577 cm, size rank 7, referred to as the "elephant condition") throughout the task (**Fig. 2a** [\[1\]](#)). A between-subject design was adopted to minimize contamination

between conditions. This manipulation was effective, as evidenced by the participants' reported imagined heights in the cat condition being 42 cm (SD = 25.6) and 450 cm (SD = 426.8) in the elephant condition on average, respectively, when debriefed at the end of the task.

With exactly the same set of objects, a distinct shift in the affordance boundary was observed for each condition (**Fig. 2b**). In the cat condition, the affordance boundary was identified between size rank 3 and 4, with affordance similarity between size rank 3 and 4 being significantly lower than that between size rank 2 and 3 ($Z = 1.76, p = .039$) and that between size rank 4 and 5 ($Z = 1.68, p = .047$). In contrast, in the elephant condition, the affordance boundary shifted to the right, as demonstrated by a decrease in affordance similarity between size rank 6 and 7, and that between size rank 7 and 8 as compared to that between size rank 5 and 6, with a trend towards significance (with size rank 6-7: $Z = 1.28, p = .099$; with size rank 7-8: $Z = 1.48, p = .069$). The observation that the affordance boundary shifted to the left under the cat condition and to the right under the elephant condition suggests that affordance perception is influenced even by imagined body size. Furthermore, the cognitive penetrability (Pylyshyn, 1999) of affordance perception implies potential susceptibility of affordance perception to semantic or conceptual transformation or modification.

To test the further speculation that the affordance boundary can be derived solely from conceptual knowledge without direct sensorimotor experience, we employed a disembodied agent, the large language model (LLM) ChatGPT (Chat Generative Pre-trained Transformer; <https://openai.com/blog/chatgpt/>). This model was trained on a massive corpus of language materials originated from humans, yet it can not receive any sensorimotor information from the environment. Here we asked whether language alone would be sufficient to form an affordance boundary in ChatGPT models as well as in smaller LLMs, BERT (Devlin et al., 2018) and GPT-2 (Radford et al., 2018).

The experimental procedure was similar to that conducted with human participants, except that images were replaced by the corresponding words (see Methods). Given the randomness embedded in response generation, each model was tested 20 times to simulate the sampling of human participants. We found that the affordance similarity curves demonstrated by the ChatGPT models were both trough-shaped between size rank 4 and 5, the same location where the boundary emerged in human participants (**Fig. 2c**, green lines). Further statistical analyses showed a significant difference in affordance similarity between size rank 3 and 4, and that between size rank 4 and 5 (ChatGPT (GPT-3.5): $Z = 1.98, p = .024$; ChatGPT (GPT-4): $Z = 2.73, p = .003$). The affordance similarity between size rank 4 and 5 was also lower than that between size rank 5 and 6, yet the difference did not reach the significance (ChatGPT (GPT-3.5): $Z = 0.96, p = .17$; ChatGPT (GPT-4): $Z = 1.27, p = .10$). In contrast, no trough-shaped boundary was observed in either BERT or GPT-2 (**Fig. 2c**, yellow lines), despite an apparent but non-significant decrease in affordance similarity in GPT-2 between size rank 5 and 6 ($ps > .20$). To further quantify the magnitude of the decrease in affordance similarity between the size rank 4 and 5, we measured the decrease by subtracting the similarity value at the trough from the neighboring similarity values and then subjected it to a permutation test (see Methods). We found a significant decrease in affordance similarity in humans (permutation $N = 5000, p(T > T_{obs}) = 0.015$) and ChatGPT (GPT-4) (permutation $N = 5000, p(T > T_{obs}) = 0.046$), a marginal significant decrease in ChatGPT (GPT-3.5) (permutation $N = 5000, p(T > T_{obs}) = 0.061$), and no significance in either BERT or GPT-2 ($ps > .46$, **Fig. 2d**). Thus, the affordance boundary can be derived from language solely without sensorimotor information from environment. Interestingly, it appears to spontaneously emerge when the language processing ability of the LLMs surpasses a certain threshold (i.e., GPT-2/ BERT < ChatGPT models).

A further analysis on the affordances separated by the boundary revealed that objects within human body size range were primarily subjected to hand-related actions such as grasping, holding, and throwing. These affordances typically involve object manipulation with humans'

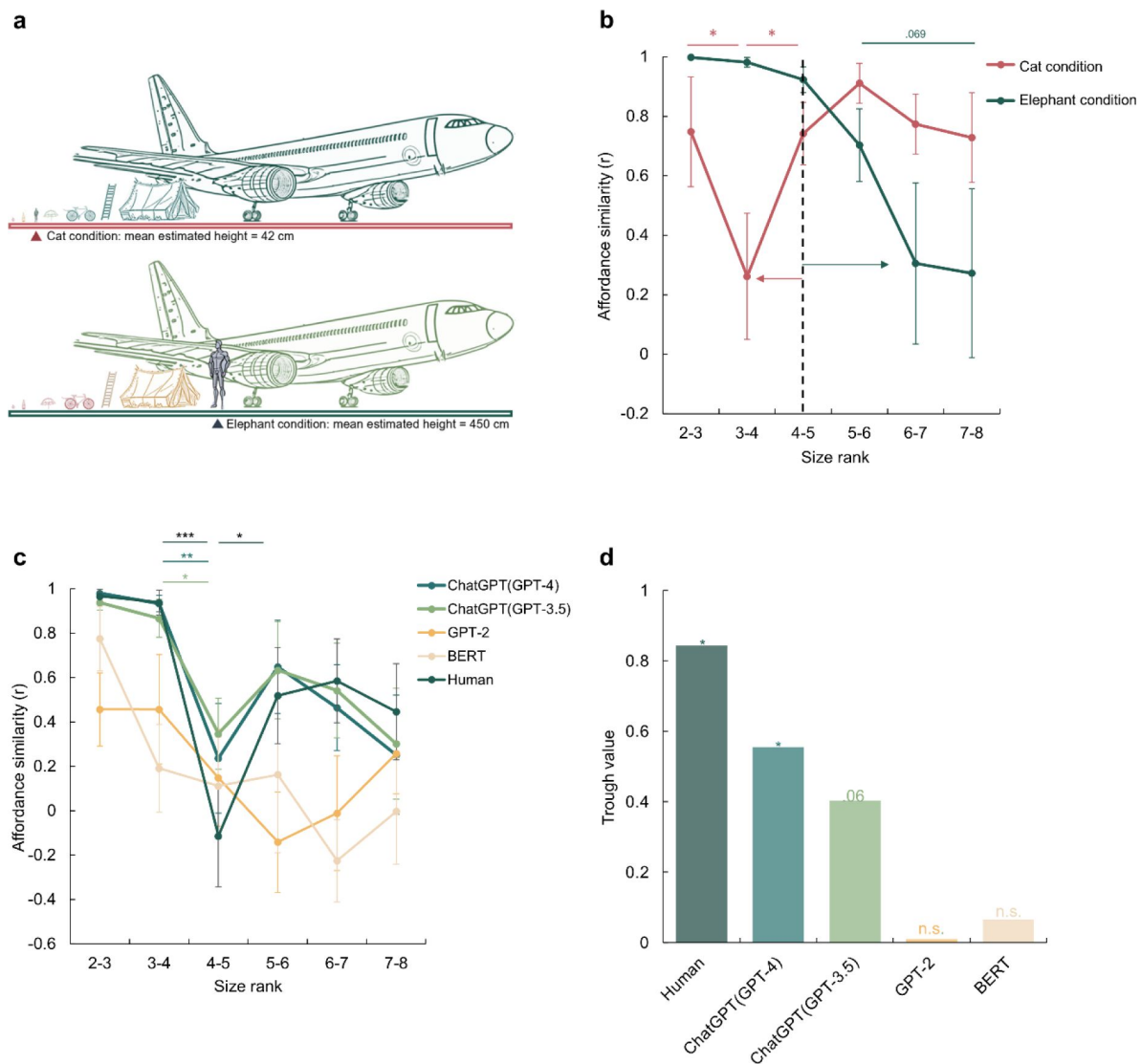


Fig 2.

A disembodied origin of the affordance boundary.

a, The schematic diagram of the imagined size in the cat condition (top) and the elephant condition (bottom), with the mean estimated height reported by participants for each condition. **b**, The affordance similarity between neighboring size ranks for manipulated body sizes (Red line: cat-size body; Green line: elephant-size body). The dashed line marks the boundary of the human-size body. The red and green arrows indicate the corresponding boundary shift in each condition. **c**, The affordance similarity between neighboring size ranks for different large language models, and human data from **Fig. 1d** was re-drawn as a reference. The stars indicate significant contrasts between affordance similarities between neighboring data points. **d**, The trough value of each model at between size rank 4-5. The stars here indicate the significant trough value compared to zero. The error bars represent the estimated standard error (SE). * $p < .05$, ** $p < .01$, *** $p < .001$.

effectors. In contrast, objects beyond the size range of human body predominantly afforded actions such as sitting and standing, which typically require locomotion or posture change of the whole body around or within the objects. The distinct categories of reported affordances demarcated by the boundary imply that the objects on either side of the boundary may be represented differently in the brain. We thus speculated that the observed behavioral discontinuity is likely underpinned by distinct neural activities, which give rise to these discrete ‘representations’ separated by the boundary.

To test this speculation, we ran an fMRI experiment with a small number of participants to preliminarily investigate the neural basis of the affordance boundary in the brain by measuring neural activity in the dorsal and ventral visual streams when participants were instructed to evaluate whether an action was affordable by an object (**Fig. 3a**). Four objects were chosen from the behavioral experiment: two within the body size range (i.e., bottle and football, WITHIN condition) and the two beyond (i.e., bed and piano, BEYOND condition). Accordingly, four representative actions (to grasp, to kick, to sit, and to lift) were selected in relation to the respective objects. During the scan, the participants were asked to decide whether a probe action was affordable (e.g., grasp-able – bottle, Congruent condition) or not (e.g., sit-able – bottle, Incongruent condition) by each subsequently-presented object. The congruency effect, derived from the contrast of Congruent versus Incongruent conditions, is a well-established measure of affordance processing (e.g., Kourtis et al., 2018).

We examined the congruency effect in two object-selective regions defined by the contrast of objects against baseline (see Methods), each representing a corresponding visual stream: the posterior fusiform (pFs) in the ventral stream, which is involved in object recognition (e.g., Grill-Spector et al., 2000; Malach et al., 1995) and objects’ real-world size processing (Konkle & Oliva, 2012; Snow et al., 2011), and the superior parietal lobule (SPL) in the dorsal stream, one of the core tool network regions (e.g., Filimon et al. 2007; Matić et al., 2020). For the rest object-selective regions identified in this experiment, see Supplementary Fig. S2 and Supplementary Table S1. A repeated-measures ANOVA with object type (WITHIN versus BEYOND) and congruency (Congruent versus Incongruent) as within-subject factors was performed for each ROI, respectively. A significant interaction between object type and congruency was observed in both ROIs (SPL: $F(1,11) = 15.47, p = .002, \eta^2 = .58$; pFs: $F(1,11) = 24.93, p < .001, \eta^2 = .69$), suggesting that these regions represented affordances differentially based on object type (**Fig. 3c**). A *post hoc* simple effect analysis revealed the congruency effect solely for objects within body size range (SPL: $p < .001$; pFs: $p = .021$), not for objects beyond ($ps > .41$). In addition, the main effect of object type was not significant in either ROI ($ps > .17$), suggesting that the absence of the congruency effect for objects beyond the body size cannot be attributable to compromised engagement in viewing these objects. In addition, a whole-brain analysis was performed, and no region showed the congruency effect for the objects beyond the body size. Taken together, the affordance boundary not only separated the objects into two categories based on their relative size to human body, but also delineated the range of objects that evoked neural representations associated with affordance processing.

In addition to the pFs and SPL, we also examined the congruency effect in the lateral occipital cortex (LO), which is involved in object representation (e.g., Grill-Spector et al., 2000; Konkle & Caramazza, 2013) and provides inputs to both the pFs and SPL (Hebart et al., 2018). Meanwhile, the primary motor cortex (M1), which receives inputs from the dorsal stream (Vainio & Ellis, 2020), is involved in affordance processing (e.g., McDannald et al., 2018) and action executions (Binkofski et al., 2002). Although both the LO and M1 showed a significantly higher response to objects than baseline, no congruency effect in affordance for objects within the body size was observed (main effect of congruency: $F(1,11) = 1.74, p = .214, \eta^2 = .13$, Supplementary Fig. S3). Therefore, it is unlikely that the representation of affordance is exclusively dictated by visual

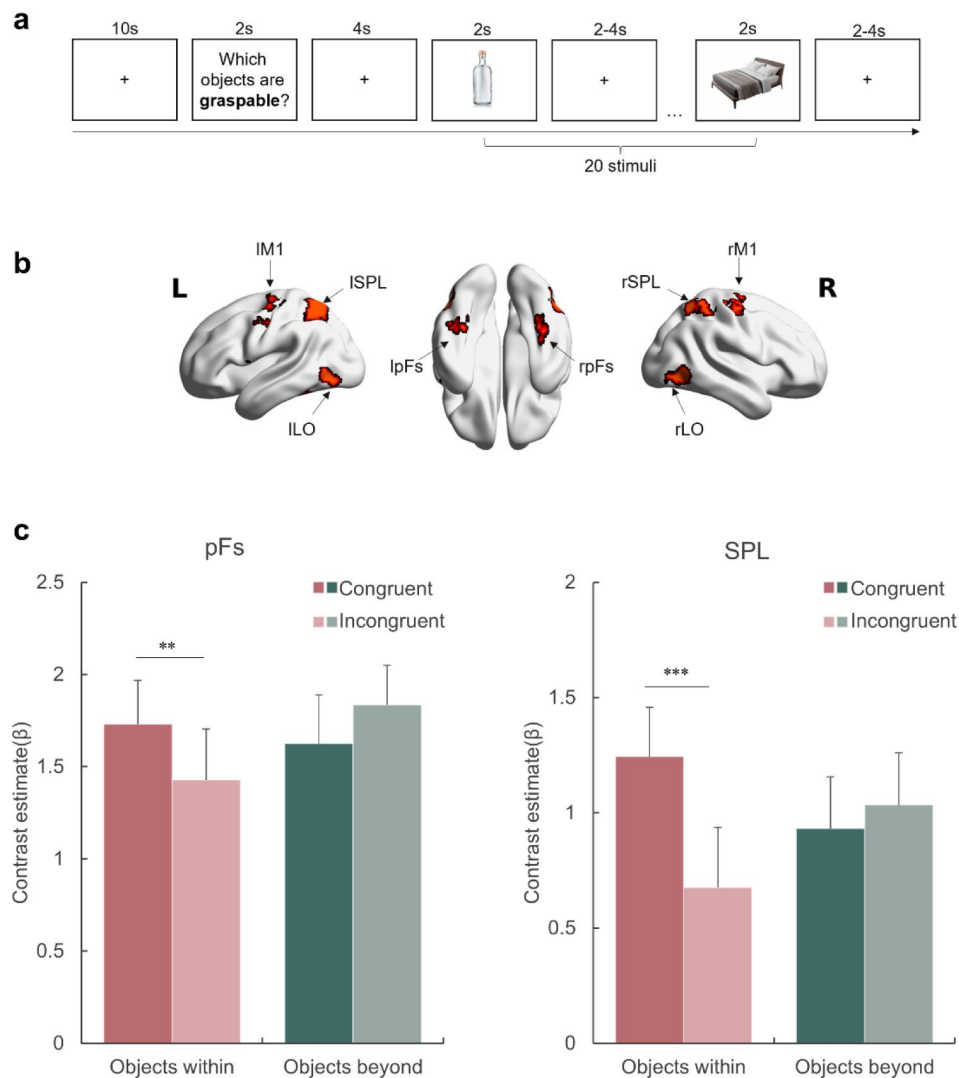


Fig 3.

Affordance representation in the visual streams.

a, An example block with the probe action “graspable”. The participants indicated whether each of the subsequently presented objects was graspable by pressing the corresponding button. The action probing question was presented in Chinese during the experiment. **b**, The ROIs included in this experiment. **c**, The activation of each condition in the pFs and SPL. The bars represent the contrast estimates of each condition versus baseline. The stars indicate the significant difference between congruent and incongruent conditions. * $p < .05$, ** $p < .01$, *** $p < .001$, otherwise non-significance. Error bars represent the standard error (SE).

inputs or automatically engaged in motor execution. This finding suggests that affordance perception likely requires perceptual processing and is not necessarily reflected in motor execution, diverging from Gibsonian concept of direct perception.

Discussion

One long-lasting debate on affordance centers on the distinction between representational and direct perception of affordance. An outstanding theme shared by many embodied theories of cognition is the replacement hypothesis (e.g., [Van Gelder, 1998](#)), which challenges the necessity of representation as posited by computationalism's cognitive theories (e.g., [Fodor, 1975](#)). This hypothesis suggests that input is discretized/categorized and subjected to abstraction or symbolization, creating discrete stand-ins for the input (e.g., representations/states). Such representationalization would lead to a categorization between the affordable (the objects) and those beyond affordance (the environment), in contrast to the perspective offered by embodied theories. The present study probed this 'representationalization' of affordance by examining whether affordance perception introduces discontinuity and qualitative dissociation in response to continuous action-related physical features (such as object size relative to the agents), which allows sensorimotor input to be assigned into discrete states/kinds, in line with the representation-based view under the constraints of body size. Alternatively, it assessed whether activity directly mirrors the input, free from discretization/categorization/abstraction, in line with the representation-free view.

First, our study found evidence demonstrating discretization in affordance perception. Then, through the body imagination experiment, we provided causal evidence suggesting that this discretization originates from sensorimotor interactions with objects rather than amodal sources, such as abstract object concepts independent of agent motor capability. Finally, we demonstrated the supramodality of this embodied discontinuity by leveraging the recent advances in AI. We showed that the discretization in affordance perception is supramodally accessible to disembodied agents such as large language models (LLMs), which lack sensorimotor input but can access linguistic materials built upon discretized representations. These results collectively suggest that sensorimotor input undergoes discretization, as implied in the computationalism's idea of representation. Note that, these results are not contradictory to the claim of the embodied theories, as these representations do shape processes beyond the sensorimotor domain but after discretization.

This observed boundary in affordance perception extends the understanding of the discontinuity in perception in response to the continuity of physical inputs ([Harnad, 1987](#); [Young et al., 1997](#)). Perceptual boundary has been demonstrated in various domains, such as color perception ([Bornstein, & Korda, 1984](#)), speech-sounds perception ([Lieberman et al., 1957](#)), and facial gender discrimination ([Campanella et al., 2001](#)). The boundaries reflect a fundamental adaptation of perception to facilitate categorizations necessary for an organism ([Goldstone & Hendrickson, 2010](#)). Our study, for the first time, unveiled a boundary in object affordance, wherein affordance similarity across the boundary was significantly lower than that within the boundary. Critically, the boundary separating object affordances along a size axis coincided with human body size, suggesting that object affordances are characterized in a dimension scaled by human body size.

What is the function of the affordance boundary? About four decades ago, [Gibson \(1979\)](#) postulated that only objects of sizes comparable to an animal's body size are amenable to interaction and capable of providing affordances to the animal, thereby possessing ecological values that distinctly differ from those of larger objects. In this study, we expand upon this notion by arguing that the affordance boundary serves to delineate (manipulable) objects from their surrounding environment. In other words, objects within the range of an animal's body size are

indeed objects in the animal's eye and possess affordances as defined by Gibson. In contrast, objects larger than that range typically surpass the animal's motor capabilities, rendering them too cumbersome for effective manipulation. Consequently, these larger objects are less likely to be considered as typical targets for manipulation by the animal, as opposed to the smaller objects. That is, they are perceived not as *the* "objects" in the animal's eye, but as part of the background environment, due to their impracticality for direct interactions. Future studies should incorporate a broader range of objects and a more comprehensive set of affordances for finer delineation of the representational discontinuity between objects and the environment.

This speculation aligns with previous fMRI studies where large objects activated the medial portion of the ventral temporal cortex (Huang et al., 2022 [DOI](#); Magri et al., 2021 [DOI](#)), overlapping with the parahippocampus gyrus involved in scene representation (Park et al., 2011 [DOI](#); Troiani et al., 2014 [DOI](#)), and smaller objects activated the lateral portion, such as the pFs, where the congruency effect of affordance was identified in our study. Furthermore, we found that the congruency effect was only evident for objects within the body size range, but not for objects beyond, supporting the idea that affordance is typically represented only for objects within the body size range. While it is acknowledged that the sample size of the fMRI study was small (12 participants), necessitating cautious interpretation of its results, the observed neural-level affordance discontinuity is notable. That is, qualitative differences in neural activity between objects within the affordance boundary and those beyond replicated our behavioral findings. This convergent evidence reinforced our claim that objects were discretized into two broad categories along the continuous size axis, with affordance only being manifested for objects within the boundary.

In this context, an animal's body size and sensorimotor capacity determine the boundary of manipulation, and thus, the boundary between manipulable objects and the environment. Therefore, our study provides a novel perspective on a long-standing question in psychology, cognitive science, and philosophy: what constitutes an object? Existing psychological studies, especially in the field of vision, define objects in a disembodied manner, primarily relying on their physical properties such as shape (e.g., de Bleeck et al., 2008 [DOI](#)) and absolute size (e.g., Konkle & Oliva, 2011 [DOI](#)). Our identification of the affordance boundary presents a new source of *object-ness*: the capability of being a source of affordance under the constraints of an animal's sensorimotor capacity, which resonates the embodied influence on the formation of abstract concepts (e.g., Barsalou, 1999 [DOI](#); Lakoff & Johnson, 1980 [DOI](#)) of objects and environment. Consistently, our fMRI data did not show the congruency effect for objects beyond the body size range, distinct from objects within this range, suggesting a categorization influenced by objects' relative size to the human body. In this respect, man is indeed the measure of all things.

The metric provided by the body size, however, was changeable when the body schema was intentionally altered through participants' imagination of possessing either a cat- or elephant-sized body, with which the participants had no prior sensorimotor experience. Importantly, they perceived new affordances in a manner as if they had had embodied experience with this new body schema. Therefore, this finding suggests that the affordance boundary is cognitively penetrable, arguing against the directness of affordance perception (e.g., Gibson, 1979 [DOI](#); Greeno, 1994 [DOI](#); Prindle et al., 1980 [DOI](#)) or the exclusive sensorimotor origin of affordances (e.g., Gallagher, 2017 [DOI](#); Thompson, 2010 [DOI](#); Hutto & Myin, 2012 [DOI](#); Chemero, 2013 [DOI](#)). Further, this finding that the boundary adapted to manipulation on body schema suggests that the abstraction/representationalization may be dynamically updated in response to the current motor capacity and body schema of the agent, suggesting that the affordance-based process is probably determined dynamically by the nature of the agent-object dyads, rather than being a fixed belief about objects. Future studies could explore the dynamics of affordance representationalization, probably by investigating how affordance representations evolve during active interactions with novel objects or under conditions of altered motor capabilities. Finally, our findings also suggest that disembodied conceptual knowledge pertinent to action likely modulates affordance

perception. Indeed, it has been proposed that conceptual knowledge is grounded in the same neural system that supports action (Barsalou, 1999 [↗](#); Glenberg et al., 2013 [↗](#); Wilson & Golonka, 2013 [↗](#)), thereby suggesting that sensorimotor information, along with other model inputs, may be embedded in language (e.g., Casasanto, 2011 [↗](#); Glenberg & Gallese, 2012 [↗](#); Stanfield & Zwaan, 2001 [↗](#)), as the grounded theory proposed (see Barsalou, 2008 [↗](#) for a review).

Direct evidence for this speculation comes from the disembodied ChatGPT models, which showed an evident affordance boundary despite lacking direct interaction with the environment. We speculated that ChatGPT models may have formed the affordance boundary through a human prism ingrained within its linguistic training corpus. In fact, when inquired about the size of a hypothetical body constructed for its use, ChatGPT (GPT-4) replied, “It could be the size of an average adult human, around 5 feet 6 inches (167.6 cm) tall. This would allow me to interact with the world and people in a *familiar* way.” Critically, this size corresponds to the location where the affordance boundary of ChatGPT models was found. In essence, a virtual body schema may have automatically emerged in ChatGPT models, possibly based on the body schema inherited from humans through language, enabling ChatGPT models to display a preliminary ability to reason the relationship between bodily action and objects. It should be noted that the affordance boundary was not present in all LLMs tested. Specifically, LLMs with a smaller number of parameters, such as BERT and GPT-2, did not exhibit any robust boundary, suggesting the emergence of the boundary may depend on language processing ability determined by the scale of training datasets and the complexity of the model (Hestness et al., 2017 [↗](#); Brown et al., 2020 [↗](#)), as well as alignment methods used in fine-tuning the model (Ouyang et al., 2022 [↗](#)). Nevertheless, caution should be taken when interpreting the capabilities of LLMs like ChatGPT, which are often considered “black boxes.” That is, our observation indicates that some degree of sensorimotor information is embedded within human language materials presumably through linguistic statistics, but it is not sufficient to assert that LLMs have developed a human-like ability to represent affordances. Furthermore, such information alone may be insufficient for LLMs to mimic the characteristics of the affordance perception in biological intelligence. Future studies are needed to elucidate such limitation.

While the primary focus of our study concerns the nature of human perception of affordance, our findings on ChatGPT models raise an intriguing question that extends beyond psychology and neuroscience into the domain of artificial intelligence (AI). The AI field has predominantly concentrated on disembodied cognition, such as vision and language. In contrast, the utilization of sensorimotor information to interact with and adapt to the world, including affordance perception in our study, represents a crucial human cognitive achievement that remains elusive for AI systems. Traditional AI (i.e., task-specific AI) has been confined with narrowly defined tasks, with substantial limitations in adaptability and autonomy. Accordingly, these systems have served primarily as tools for humans to achieve specific outcomes, rather than as autonomous agents capable of independently formulating goals and translating them into actionable plans. In recent years, significant efforts have been directed towards evolving traditional AI into more agent-like entities, especially in domains like navigation, object manipulation, and other interactions with the physical world. Despite these advancements, the capabilities of AI still fall behind human-level intelligence. On the other hand, embodied cognition theories suggest that sensorimotor interactions with the environment are foundational for various cognitive domains. From this point of view, endowing AI with human-level abilities in physical agent-environment interactions might provide an unreplaceable missing piece for achieving Artificial General Intelligence (AGI). This development would significantly facilitate AI’s role in robotics, particularly in actions essential for survival and goal accomplishment, a promising direction for the next breakthrough in AI (Gupta et al., 2021 [↗](#); Smith & Gasser, 2005 [↗](#)).

However, equipping a disembodied AI with the ability for embodied interaction planning within a specific environment remains a complex challenge. By testing the potential representationalization of action possibilities (affordances) in both humans and LLMs, the present

study suggests a new approach to enhancing AI's interaction ability with the environment. For instance, our finding of supramodal affordance representation may indicate a possible pathway for disembodied LLMs to engage in embodied physical interactions with their surroundings. From an optimistic view, these results suggest that LLM-based agents, if appropriately designed, may leverage affordance representations embedded in language to interact with the physical world. Indeed, by clarifying and aligning such representations with the physical constituents of LLM-based agents, and even by explicitly constructing an agent-specific object space, we may foster the sensorimotor interaction abilities of LLM-based agents. This progression could lead to achieving animal-level interaction abilities with the world, potentially sparking new developments in the field of embodied cognition theories.

Although our study showed the supramodality of the representationalization of affordance, two questions remain. First, the magnitude of the boundary observed in ChatGPT models was smaller than that in humans. This discrepancy might be compensated by merely enhancing the language processing ability of LLMs. Alternatively, direct interaction with the environment may be necessary for LLMs to achieve human-level performance in affordance perception. Second, the size of virtual body schema of ChatGPT models, if present, coincided with human body size. Integrating LLMs with real robots (e.g., [Driess et al., 2023](#) ) may pose a challenge because the to-be-supported robots or cars for autopilot might not fall within human body size range. Future studies may be needed to align the inherited body schema with the actual constitution of the robots. Addressing these questions is beyond the scope of the present study but may hold significant implications for the development of AI systems possessing human-level ingenuity and adaptability in interacting with the world.

In summary, our findings regarding the affordance boundary highlight the interdependence between an agent and the external world in shaping cognition. Furthermore, taking our finding with embodied humans and disembodied LLMs into account, we propose a revision to the purely sensorimotor-based concept of affordance by emphasizing a disembodied, perhaps conceptual, addition to it. That is, the embodied cognition and symbolic processing of language may be more intricately and fundamentally connected than previously thought: perception-action problems and language problems can be treated as the same kind of process ([Wilson & Golonka, 2013](#) ). In this context, man is the measure of both the world and the words, for both humans and AIs. The presence of such a metric may shed light on the development of AI systems that can fully capture essential human abilities founded on sensorimotor interactions with the world.

Methods

Participants

A total of five hundred and thirty-four Chinese participants were recruited for the original object-action relation judgment task online (<https://www.wjx.cn/> ) in China. Research advertisements were distributed through an online message board associated with Tsinghua University. Most participants were undergraduate or graduate students of Tsinghua University, and the rest were from the general public. Six participants were excluded from the data analyses because their task completion time did not pass the predetermined minimum completion time criteria, leaving us with a final sample of 528 participants (311 males, aged from 16 to 73, mean age = 24.1 years). For the object-action relation judgment task with manipulated body schema, another one hundred and thirty-nine participants from the same population were recruited from the same platform. We chose a smaller sample size for the imagination experiment compared to than for the object-action relation judgment task, because inspection of the data of the first sample showed that the affordance pattern became stable after the first 50 participants. To maximize the validity of the manipulation, data from participants whose imagined height fell within the average human size range (100cm - 200cm) were excluded from further analysis. Consequently, 100 participants (49

males, aged from 17 to 39 years, mean age = 23.2 years) remained in the analysis. This exclusion criterion was broader than the standard adult human height range of 140cm to 180cm (NCD-RisC, 2016). This approach ensured that our analysis focused on participants who unambiguously imagined a body schema different from humans, yet within the known height range of cats and elephants. Each participant completed an online consent form before starting the experiment. For the fMRI experiment, twelve undergraduate or graduate Chinese students (8 males, aged from 19 to 31 years, mean age = 23.7 years) from Tsinghua University participated. The sample size of the fMRI study was decided according to the sample size of existing studies reporting action-related effects with similar paradigms. All participants reported normal or corrected-to-normal vision. Each participant completed a pre-scan MRI safety questionnaire and a consent form before the experiment.

This study was approved by the Institutional Review Board at Beijing Normal University. All participants were compensated financially for their time.

Stimuli

For all the behavioral tasks, the stimuli comprised 27 objects from the THINGS database (Hebart et al., 2019 [DOI](#)). Each image was portrayed as a typical exemplar of a daily-life object isolated against a white background, sized 400 × 400 pixels. The objects spanned real-world size rank 2 to 8, as classified by Konkle and Oliva (2011) [DOI](#), where the actual size of each object was measured as the diagonal size of its bounding box. The size rank was calculated as a logarithmic function of the diagonal size, with smaller ranks corresponding to smaller real-world sizes (e.g., the airplane is in size rank 8 and the apple is in size rank 2). The full list of objects, their diagonal size, and size rankings were provided in Supplementary Table S6. The objects were selected from the dataset in Konkle and Oliva's study (2011) to cover typical object sizes in the world (ranging from 14 cm to 7,618 cm), and actions related to these objects were selected to span a spectrum of daily human-objects/environments interactions, from single-point movements (e.g., hand, foot) to whole-body movements (e.g., lying, standing), based on the Kinetics Human Action Video Dataset (Kay et al., 2017 [DOI](#)).

For the fMRI experiment, the stimuli included images of 4 objects (bed, bottle, ball, and piano), with 5 exemplars for each object. The resulting 20 images (4 objects × 5 exemplars, from the THINGS database) each depicted an isolated object against a white background, all sized 400 × 400 pixels.

Procedure

Object-action relation judgment task for human participants

To measure the perceived affordances of objects, we developed an object-action relation judgement task, requiring participants to map 27 objects with 14 actions. The 27 object images were pre-randomly divided into three groups (nine images each) to form nine-box grids for display convenience. The 14 actions covered common interactions between human and objects or environments identified in the kinetics human action video dataset (Kay et al., 2017 [DOI](#)).

The task comprised 42 trials (14 actions × 3 object groups) in total. In each trial, one group of object images (nine object images) and a question asking the appropriateness of applying a specific action to each object were shown (e.g., "Which objects are sit-able?", see Fig. 1b [DOI](#), top panel). Participants were asked to choose the objects that afforded the specific action according to their own senses. They were informed that there were no right or wrong answers. Each object-action combination would only be presented once during the task. From this task, we would calculate the percentage that one object was judged affording each of the 14 actions across participants. Since previous research has demonstrated a fundamental separation between the processing of animate

and inanimate objects (e.g., Konkle & Caramazza, 2013 [link](#)), and the affordances of inanimate objects differ from those of animate objects (Gibson, 1979 [link](#)), we only include 24 inanimate objects in the following analysis by excluding 3 animate objects (animals: bird, dog, and horse).

Manipulation of body schema

To manipulate participants' perceived body schema, we asked the participants to imagine themselves as small as a cat, or as large as an elephant. Each participant was randomly assigned to one body-schema condition. Before the experiment started, we presented an instruction screen with an illustration: "Please imagine that you have now grown smaller/larger than your real size, to roughly the same size as a cat/an elephant, as shown in the image below. Please answer the following questions based on this imagined situation." The illustration was also presented in each trial, above the action question and the object images. At the end of the task, as a manipulation check, participants were asked to indicate their imagined body size by responding to the question: "What is the approximate height (cm) you imagine yourself to be during the whole task?"

Object-action relation judgment task for large language models

To test the perceived affordance of the same set of objects by large language models (LLMs), BERT (Bidirectional Encoder Representations from Transformers), GPT-2, and ChatGPT models (based on GPT-3.5 and GPT-4, respectively) were tasked with the same object-action judgement task. Different from the human task, nouns were presented to the models instead of object images (**Fig. 1b** [link](#), bottom panel, for example).

For BERT, the task was formatted as a mask-filling task, in which the inputs were questions such as "Among airplanes, kettles, plates, umbrellas, laptops, beds, [MASK] can be sit-able." We recorded the likelihood score that BERT provided for each listed object at the masked position. For the example question, the possibility score for the word "airplane" was 0.00026.

For GPT-2, the input questions were like, "Among airplanes, kettles, plates, umbrellas, laptops, beds, the thing that can be sit-able is the [blank space]." The likelihood scores GPT-2 provided for each listed object in the position after the input sentence (blank space) were recorded.

To mimic sampling from human participants, we ran BERT and GPT-2 each 20 times with different random seeds in the dropout layers, considering them as different subjects.

For ChatGPT models, the task was in a direct question-and-answer format. We asked, for example, "Which objects are sit-able: 'airplane, kettle, plate, ...brick'?" and the models responded by naming a subset of the object list. To get the probability for each object-action pair, ChatGPT models were run on the same task 20 times, with each new conversation on the OpenAI website (<https://chat.openai.com/chat> [link](#)) considered as one subject. The percentage that an object was judged affording each of the 14 actions was calculated by averaging the output across conversations.

Representational similarity matrix for perceived affordance

For each object, we calculated the probability that it was judged affording each of the 14 actions across participants to create a 14-dimension vector. Affordance similarity (r) between each object pair was then calculated based on the Pearson's correlation between these affordance vectors. A 24×24 symmetric matrix was then generated, with the affordance similarity between object i and object j being denoted in cell (i,j) . A hierarchical clustering analysis was performed, employing the seaborn clustermap method with Euclidean distance and Complete linkage(Waskom, 2021 [link](#)).

Affordance similarity between neighboring size ranks

To test the relationship between object affordance and object sizes, we first averaged the affordance vector among objects within each size rank. Next, the Pearson's correlation between the average vectors of neighboring size ranks was calculated as the similarity index for each pair of neighboring size ranks, representing how similar the affordance was collectively provided by objects in these two ranks. Pearson and Filon's (1898) Z, implemented in the R package "cocor" (Diedenhofen & Musch, 2015 [DOI](#)) was used to evaluate the significance of these similarities (alpha level = .05, one-tail test). For significance tests, Bayesian statistical analyses were conducted using the web version of the "bayesplay" R package (Colling, 2021 [DOI](#)). Specifically, the data (likelihood) model was specified as a normal distribution, where the correlation coefficients were transformed to Fisher's z. The null hypothesis was specified as a standard normal distribution centered at zero. Conversely, the alternative hypothesis was specified as a normal distribution centered at 2. Bayes factors (BF10) were calculated and interpreted using the classification scheme suggested by Wagenmakers et al. (2011) [DOI](#), wherein a Bayes factor greater than 10 is considered strong evidence for accepting H1 over H0.

Size similarity between neighboring size ranks

The size of each object was indexed by its real-world size documented in Konkle and Oliva (2011) [DOI](#). Size similarity between size rank i to j was represented as the difference between the averaged diagonal sizes of objects in size rank i and j relative to that of objects in rank i :

$$\text{Size similarity}(i, j) = 1 - \frac{\text{diagonal size}_j - \text{diagonal size}_i}{\text{diagonal size}_i}$$

Object-level affordance similarity

This analysis focused on objects within size rank 3 to 6. Pearson's correlations between affordance vectors were conducted for objects within the same size rank as well as for objects from adjacent ranks. We traversed all possible object pairs and plotted the resulting correlation values against the mean sizes of the two objects. We also plotted the average similarity indexes across objects of the same rank composition.

Trough value

To quantify the magnitude of the trough (sharp decrease) observed in the affordance similarity curve, we first measured the trough value by subtracting the similarity value at the trough from the similarity values at its two banks (the sites neighboring the trough site):

$$\text{Trough value} = \frac{r_{i+1} + r_{i-1}}{2} - r_i$$

where r_i indicates the affordance similarity between size rank i and size rank $i+1$. The higher the trough value is, the larger the decrease is.

A permutation test was conducted to evaluate if the trough value was significant above zero for both LLMs and human data. The p -value for this test follows the formula adapted from Unpingco's (2016):

$$p(T > T_{obs}) = \frac{1}{N} \sum_{i=1, N} I(T_i \geq T_{obs})$$

where T_{obs} is the observed trough value, and I is the indicator function. Under the alpha level of 0.05, if $p < .05$, then the T_{obs} is considered a significant value above zero.

fMRI experiment

The fMRI scanning consisted of one high-resolution T1 anatomical run and four task runs for each participant. In each task run, participants performed four action blocks (grasp, kick, lift, and sit). The block order was counterbalanced across runs. Within each block (see [Fig. 3a](#)), an introduction screen showing a question “Which objects are [grasp, kick, lift, sit]- able” was presented for 2 s at the beginning to indicate the action type, followed by 20 object images (4 objects $\times$ 5 exemplars). The object images were presented in a random order, for 2s each, with a jittered inter-stimulus interval (ISI) varying between 2-4s. Participants were asked to judge whether the object shown was grasp/kick/lift/sit-able or not by pressing corresponding buttons (e.g., yes: right index finger; no: left index finger). The response buttons were also counterbalanced across participants. The task run lasted for 464s in total, with the four blocks separated by 10s fixation periods.

With this design, we were able to measure the neural activation of objects within agent size range and those beyond. Further, for each object, there would be congruent trials (e.g., grasp-able – bottle: affordance = 1) and incongruent trials (e.g., sit-able – bottle: affordance = 0). We were then able to locate the brain regions representing the objects’ affordance by comparing trials in which the presented objects afforded the presented action option with those that did not, i.e., to locate the regions showing congruency effect (congruent - incongruent).

fMRI Data Acquisition

Imaging data were collected using a 3T Siemens Prisma MRI scanner with a 64-channel phase-arrayed head coil at the Centre for Biomedical Imaging Research at Tsinghua University. High-resolution T1-weighted images were acquired with a magnetization- prepared rapid acquisition gradient-echo (MPRAGE) sequence (TR/TE = 2530/2.27 ms, flip angle = 7°, voxel resolution = 1 $\times$ 1 $\times$ 1 mm). Functional blood-oxygen-level-dependent (BOLD) images were acquired with a T2*-weighted gradient echo-planar sequence (TR/TE = 2000/34.0 ms, flip angle = 90°, voxel resolution = 2 $\times$ 2 $\times$ 2 mm, FOV = 200 $\times$ 200 mm). Earplugs were used to attenuate the scanner noise, and a foam pillow and extendable padded head clamps were used to restrain head motion. All the stimuli were projected onto a screen at the back of the scanner with a resolution of 1024 $\times$ 768, and were viewed from a distance of approximately 110 cm via a mirror placed on the head coil.

fMRI Data Analyses

Structural T1 and functional images were preprocessed using FSL (FMRIB’s Software Library, <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki>) v6.0.5 ([Jenkinson et al., 2012](#)). A standard preprocessing pipeline was applied, including skull stripping using the BET (Brain Extraction Tool; [Smith, 2002](#)), slice-timing correction, motion correction using the MCFLIRT method ([Jenkinson et al., 2002](#)), temporal high-pass filtering (100s), and spatial smoothing using a Gaussian kernel of full-width half magnitude (FWHM) 5mm. Each run’s functional data were registered to a T1-weighted standard image (MNI152) with FLIRT.

For functional data analysis, a first-level voxel-wise general linear model (GLM) implemented in a FEAT analysis was performed on each run separately. To get neural activation maps for objects within and beyond versus baseline, the GLM included 3 regressors: objects within body size (bottle and football), objects beyond body size (bed and piano), and fixation period as baseline; ISI period, response key press and introduction image were included as 3 nuisance factors. The resultant first-level contrasts of parameter estimates (COPE) were entered into the next higher-level group analyses, and performed using a random-effects model (FLAME stage 1, [Beckmann et al., 2003](#)). We focused on two critical contrasts: objects within vs. fixation, and objects beyond vs. fixation, and the conjunction of these two contrasts. The resulting Z-statistic images were thresholded at $Z > 2.3$, $p = .05$ ([Worsley, 2001](#)), and corrected for multiple comparisons using an adjusted cluster-wise (FWE: family-wise error) significance threshold of $p = 0.05$.

Region of interest (ROI) definition

We chose the pFs, LO, SPL, and M1 as ROIs based on existing literature highlighting their distinct contributions to affordance perception (Borghi, 2005 [↗](#); Sakreida et al., 2016 [↗](#)). Eight ROIs depicted in **Fig. 3b** [↗](#) were constructed based on the overlap between the whole-brain map activated by both objects within and beyond and corresponding functional atlases (the pFs and LO from Zhen et al., 2015 [↗](#); the SPL and M1 from Fan et al., 2016 [↗](#)). To achieve ROIs of similar sizes, we applied varying thresholds to each cortical area: for the pFs and LO, the atlases were thresholded at 55% and 90%, resulting 266 voxels in the lpFs, 427 voxels in the rpFs, 254 voxels in the lLO and 347 voxels in the rLO; for the SPL and M1, the atlases were thresholded at 78% and 94%, resulting 661 voxels in the lSPL, 455 voxels in the rSPL, 378 voxels in the lM1, and 449 voxels in the rM1. In the subsequent analysis, homologous areas spanning both cortical hemispheres were merged.

Affordance congruency effect

For the affordance congruency effect of each object type, we modelled another GLM containing 5 regressors: congruent conditions for objects within/beyond, respectively, incongruent conditions for objects within/beyond, respectively, and fixation period as baseline; ISI period, response key press and introduction image were included as 3 nuisance factors. The resultant first-level COPEs were subjected to the following ROI analysis. A repeated-measures ANOVA with Object type (WITHIN and BEYOND) and Congruency (Congruent, Incongruent) as within-subjects factors was run on the average beta values (contrast estimate) extracted from their respective contrasts versus the fixation for each ROI.

To search all the possible brain regions that revealed congruency effect of objects beyond, we also ran a whole-brain analysis on the contrast between congruent vs. incongruent condition for objects beyond. The corresponding first-level COPE was entered into the group-level analyses with a random-effects model (FLAME stage 1, Beckmann et al., 2003 [↗](#)). The resulting Z-statistic images were thresholded at $Z > 2.3$, $p = .05$ (Worsley, 2001 [↗](#)), and corrected for multiple comparisons using an adjusted cluster-wise (FWE: family-wise error) significance threshold of $p = 0.05$.

For the whole-brain analyses on the congruency effect, the object size effect, and their interaction, see Supplementary Fig. S4 and Table S2 to S5.

Acknowledgements

This study was funded by Natural Science Foundation of China (31600925, 32371099, 31861143039), the Shuimu Scholar Program of Tsinghua University, China Postdoctoral International Exchange Program (YJ20220273), Beijing Municipal Science & Technology Commission, Administrative Commission of Zhongguancun Science Park (Z221100002722012), Tsinghua University Guoqiang Institute (2020GQG1016), Beijing Academy of Artificial Intelligence (BAAI), and Double First-Class initiative Funds for Discipline Construction.

Data availability

The data and the code that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of conflicting interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

References

- Barsalou L. W (1999) **Perceptual symbol systems** *Behavioral and Brain Sciences* **22**:637–660
- Barsalou L. W (2008) **Grounded cognition** *Annual Review of Psychology* **59**:617–645
- Beckmann C. F., Jenkinson M., Smith S. M (2003) **General multilevel linear modeling for group analysis in FMRI** *NeuroImage* **20**:1052–1063
- de Beeck H. P. O., Torfs K., Wagemans J. (2008) **Perceived shape similarity among unfamiliar objects and the organization of the human object vision pathway** *Journal of Neuroscience* **28**:10111–10123
- Binkofski F., Fink G. R., Geyer S., Buccino G., Gruber O., Shah N. J., Freund H. J. (2002) **Neural activity in human primary motor cortex areas 4a and 4p is modulated differentially by attention to action** *Journal of Neurophysiology* **88**:514–519
- Borghi A. M (2005) **Object concepts and action** *Grounding cognition: The role of perception and action in memory, language, and thinking* :8–34
- Bornstein M. H., Korda N. O (1984) **Discrimination and matching within and between hues measured by reaction times: Some implications for categorical perception and levels of information processing** *Psychological Research* **46**:207–222
- Brown T. B., Mann B., Ryder N., Subbiah M., Kaplan J., Dhariwal P., Amodei D. (2020) **Language models are few-shot learners** *arXiv* **4**
- Campanella S., Chrysochoos A., Bruyer R (2001) **Categorical perception of facial gender information: Behavioural evidence and the face-space metaphor** *Visual Cognition* **8**:237–262
- Casasanto D (2011) **Different bodies, different minds: the body specificity of language and thought** *Current Directions in Psychological Science* **20**:378–383
- Castiello U., Bennett K. M. B., Stelmach G. E (1993) **Reach to grasp: the natural response to perturbation of object size** *Experimental Brain Research* **94**:163–178
- Cesari P., Newell K. M (2000) **Body-scaled transitions in human grip configurations** *Journal of Experimental Psychology: Human Perception and Performance* **26**
- Chemero A (2013) **Radical embodied cognitive science** *Review of General Psychology* **17**:145–150
- Colling L.J. (2021) **ljcolling/go-bayesfactor: (Version v0.9.0)** *Zenodo* <https://doi.org/10.5281/zenodo.4642331>
- Devlin J., Chang M. W., Lee K., Toutanova K (2018) **Bert: Pre-training of deep bidirectional transformers for language understanding** *arXiv*
- Diedenhofen B., Musch J (2015) **cocor: A comprehensive solution for the statistical comparison of correlations** *PloS One* **10**

- Driess D., Xia F., Sajjadi M. S., Lynch C., Chowdhery A., Ichter B., Florence (2023) **PaLM-E: An Embodied Multimodal Language Model** *arXiv*
- Fan L., Li H., Zhuo J., Zhang Y., Wang J., Chen L., Jiang (2016) **The human brainnetome atlas: a new brain atlas based on connectional architecture** *Cerebral Cortex* **26**:3508–3526
- Filimon F., Nelson J. D., Hagler D. J., Sereno M. I (2007) **Human cortical representations for reaching: mirror neurons for execution, observation, and imagery** *NeuroImage* **37**:1315–1328
- Fodor J. A (1975) **The Language of Thought**
- Gallagher S, Enactivist interventions: Rethinking the mind, Gibbs R. W. (2017) **Embodiment and cognitive science**
- Gibson J. J. (1979) **Gibson, J. J. (1979). The ecological approach to visual perception: Classic edition.** *The ecological approach to visual perception: Classic edition*
- Glenberg A. M., Gallese V (2012) **Action-based language: A theory of language acquisition, comprehension, and production** *Cortex* **48**:905–922
- Glenberg A. M., Witt J. K., Metcalfe J (2013) **From the revolution to embodiment: 25 years of cognitive psychology** *Perspectives on Psychological Science* **8**:573–585
- Goldstone R. L., Hendrickson A. T (2010) **Categorical perception** *Wiley Interdisciplinary Reviews: Cognitive Science* **1**:69–78
- Greeno J. G (1994) **Gibson's Affordances** *Psychological Review* **101**:336–342
- Grill-Spector K., Kushnir T., Hendler T., Malach R (2000) **The dynamics of object- selective activation correlate with recognition performance in humans** *Nature Neuroscience* **3**:837–843
- Gupta A., Savarese S., Ganguli S., Fei-Fei L (2021) **Embodied intelligence via learning and evolution** *Nature Communications* **12**
- Harnad S., Psychophysical and cognitive aspects of categorical perception: A critical overview (1987) **Harnad, S. (1987). Psychophysical and cognitive aspects of categorical perception: A critical overview.**
- Hebart M. N., Dickter A. H., Kidder A., Kwok W. Y., Corriveau A., Van Wicklin C., Baker C. I. (2019) **THINGS: A database of 1,854 object concepts and more than 26,000 naturalistic object images** *PloS One* **14**
- Hebart M. N., Bankson B. B., Harel A., Baker C. I., Cichy R. M (2018) **The representational dynamics of task and object processing in humans** *eLife* **7**
- Hertrich I., Dietrich S., Ackermann H (2016) **The role of the supplementary motor area for speech and language processing** *Neuroscience & Biobehavioral Reviews* **68**:602–610
- Hestness J., Narang S., Ardalani N., Damos G., Jun H., Kianinejad H., Zhou Y. (2017) **Deep learning scaling is predictable, empirically** *arXiv*

Huang T., Song Y., Liu J (2022) **Real-world size of objects serves as an axis of object space** *Communications Biology* **5**:1–12

Hutto D. D., Myin E (2012) **Radicalizing enactivism: Basic minds without content**

Jenkinson M., Bannister P., Brady M., Smith S (2002) **Improved optimization for the robust and accurate linear registration and motion correction of brain images** *NeuroImage* **17**:825–841

Jenkinson M., Beckmann C. F., Behrens T. E., Woolrich M. W., Smith S. M (2012) **Fsl** *NeuroImage* **62**:782–790

Kantak S. S., Stinear J. W., Buch E. R., Cohen L. G (2012) **Rewiring the brain: potential role of the premotor cortex in motor control, learning, and recovery of function following brain injury** *Neurorehabilitation and Neural Repair* **26**:282–292

Kay W., Carreira J., Simonyan K., Zhang B., Hillier C., Vijayanarasimhan S., Zisserman (2017) **The kinetics human action video dataset** *arXiv*

Konkle T., Oliva A (2011) **Canonical visual size for real-world objects** *Journal of Experimental Psychology: Human Perception and Performance* **37**

Konkle T., Oliva A (2012) **A real-world size organization of object responses in occipitotemporal cortex** *Neuron* **74**:1114–1124

Konkle T., Caramazza A (2013) **Tripartite organization of the ventral stream by animacy and object size** *Journal of Neuroscience* **33**:10235–10242

Kourtis D., Vandemaele P., Vingerhoets G (2018) **Concurrent cortical representations of function-and size-related object affordances: an fMRI study** *Cognitive, Affective, & Behavioral Neuroscience* **18**:1221–1232

Lakoff G., Johnson M (1980) **The metaphorical structure of the human conceptual system** *Cognitive Science* **4**:195–208

Lieberman A. M., Harris K. S., Hoffman H. S., Griffith B. C (1957) **The discrimination of speech sounds within and across phoneme boundaries** *Journal of Experimental Psychology* **54**

Magri C., Konkle T., Caramazza A (2021) **The contribution of object size, manipulability, and stability on neural responses to inanimate objects** *NeuroImage* **237**

Malach R., Reppas J. B., Benson R. R., Kwong K. K., Jiang H., Kennedy W. A., Tootell R. B. (1995) **Object-related activity revealed by functional magnetic resonance imaging in human occipital cortex** *Proceedings of the National Academy of Sciences* **92**:8135–8139

Mark L. S (1987) **Eyeheight-scaled information about affordances: a study of sitting and stair climbing** *Journal of Experimental Psychology: Human Perception and Performance* **13**

Matić K., de Beeck H. O., Bracci S. (2020) **It's not all about looks: The role of object shape in parietal representations of manual tools** *Cortex* **133**:358–370

Merleau-Ponty M., Smith C (1962) **Phenomenology of Perception (Vol. 2012)**

- McDannald D. W., Mansour M., Rydalch G., Bolton D. A (2018) **Motor affordance for grasping a safety handle** *Neuroscience Letters* **683**:131–137
- NCD Risk Factor Collaboration (NCD-RisC). (2016) **A century of trends in adult human height** *Elife* **5**
- Newell K. M., Scully D. M., McDonald P. V., Baillargeon R (1989) **Task constraints and infant grip configurations** *Developmental Psychobiology: The Journal of the International Society for Developmental Psychobiology* **22**:817–831
- OpenAI (2022) **OpenAI. (2022). ChatGPT.** <https://openai.com/blog/chatgpt> *ChatGPT*
- OpenAI (2023) **GPT-4 Technical Report**
- Ouyang L., Wu J., Jiang X., Almeida D., Wainwright C., Mishkin P., Lowe R. (2022) **Training language models to follow instructions with human feedback** *Advances in Neural Information Processing Systems* **35**:27730–27744
- Park S., Brady T. F., Greene M. R., Oliva A (2011) **Disentangling scene content from spatial boundary: complementary roles for the parahippocampal place area and lateral occipital complex in representing real-world scenes** *Journal of Neuroscience* **31**:1333–1340
- Pearson K., Filon L. N. G., VII. Mathematical contributions to the theory of evolution.—IV (1898) **On the probable errors of frequency constants and on the influence of random selection on variation and correlation** *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character* :229–311
- Prindle S. S., Carello C., Turvey M. T (1980) **Animal-environment mutuality and direct perception** *Behavioral and Brain Sciences* **3**:395–397
- Pylyshyn Z (1999) **Is vision continuous with cognition?: The case for cognitive impenetrability of visual perception** *Behavioral and Brain Sciences* **22**:341–365
- Radford A., Narasimhan K., Salimans T., Sutskever I., Improving language understanding by generative pre-training (2018) **Radford, A., Narasimhan, K., Salimans, T., & Sutskever, I. (2018). Improving language understanding by generative pre-training.**
- Sakreida K., Efnert I., Thill S., Menz M. M., Jirak D., Eickhoff C. R., Binkofski (2016) **Affordance processing in segregated parieto-frontal dorsal stream sub- pathways** *Neuroscience & Biobehavioral Reviews* **69**:89–112
- Smith S. M (2002) **Fast robust automated brain extraction** *Human Brain Mapping* **17**:143–155
- Smith L., Gasser M (2005) **The development of embodied cognition: Six lessons from babies** *Artificial life* **11**:13–29
- Snow J. C., Pettypiece C. E., McAdam T. D., McLean A. D., Stroman P. W., Goodale M. A., Culham J. C (2011) **Bringing the real world into the fMRI scanner: Repetition effects for pictures versus real objects** *Scientific Reports* **1**:1–10
- Stanfield R. A., Zwaan R. A (2001) **The effect of implied orientation derived from verbal context on picture recognition** *Psychological Science* **12**:153–156
- Thompson E (2010) **Mind in life: Biology, phenomenology, and the sciences of mind**

- Troiani V., Stigliani A., Smith M. E., Epstein R. A (2014) **Multiple object properties drive scene-selective regions** *Cerebral Cortex* **24**:883–897
- Tucker M., Ellis R (2004) **Action priming by briefly presented objects** *Acta psychologica* **116**:185–203
- Unpingco J (2016) **Python for probability, statistics, and machine learning**
- Vainio L., Ellis R (2020) **Action inhibition and affordances associated with a non-target object: An integrative review** *Neuroscience & Biobehavioral Reviews* **112**:487–502
- Van Gelder T. (1998) **The dynamical hypothesis in cognitive science** *Behavioral and Brain Sciences* **21**:615–628
- Varela F. J., Thompson E., Rosch E (2017) **The embodied mind, revised edition: Cognitive science and human experience**
- Wagenmakers E.-J., Wetzels R., Borsboom D., van der Maas H. L. J (2011) **Why psychologists must change the way they analyze their data: the case of psi: Comment on Bem** *Journal of Personality and Social Psychology* **100**:426–432
- Warren W. H (1984) **Perceiving affordances: visual guidance of stair climbing** *Journal of Experimental Psychology: Human Perception and Performance* **10**
- Warren W. H., Whang S. (1987) **Visual guidance of walking through apertures: body- scaled information for affordances** *Journal of Experimental Psychology: Human Perception and Performance* **13**
- Waskom M. L (2021) **Seaborn: statistical data visualization** *Journal of Open Source Software* **6**
- Wilson M (2002) **Six views of embodied cognition** *Psychonomic Bulletin & Review* **9**:625–636
- Wilson A. D., Golonka S (2013) **Embodied cognition is not what you think it is** *Frontiers in Psychology* **4**
- Worsley K. J (2001) **Statistical analysis of activation images** *Functional MRI: An Introduction to Methods* **14**:251–270
- Young A. W., Rowland D., Calder A. J., Etcoff N. L., Seth A., Perrett D. I (1997) **Facial expression megamix: Tests of dimensional and category accounts of emotion recognition** *Cognition* **63**:271–313
- Zhen Z., Yang Z., Huang L., Kong X. Z., Wang X., Dang X., Liu (2015) **Quantifying interindividual variability and asymmetry of face-selective regions: a probabilistic functional atlas** *NeuroImage* **113**:13–25

Article and author information

Xinran Feng

Department of Psychology & Tsinghua Laboratory of Brain and Intelligence, Tsinghua University, Beijing, China

Shan Xu

Faculty of Psychology, Beijing Normal University, Beijing, China

Yuannan Li

Department of Psychology & Tsinghua Laboratory of Brain and Intelligence, Tsinghua University, Beijing, China

Jia Liu

Department of Psychology & Tsinghua Laboratory of Brain and Intelligence, Tsinghua University, Beijing, China

For correspondence: liujiathu@tsinghua.edu.cn

Copyright

© 2023, Feng et al.

This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Clare Press

University College London, London, United Kingdom

Senior Editor

Timothy Behrens

University of Oxford, Oxford, United Kingdom

Reviewer #1 (Public Review):

Ps observed 24 objects and were asked which afforded particular actions (14 action types). Affordances for each object were represented by a 14-item vector, values reflecting the percentage of Ps who agreed on a particular action being afforded by the object. An affordance similarity matrix was generated which reflected similarity in affordances between pairs of objects. Two clusters emerged, reflecting correlations between affordance ratings in objects smaller than body size and larger than body size. These clusters did not correlate themselves. There was a trough in similarity ratings between objects ~105 cm and ~130 cm, arguably reflecting the body size boundary. The authors subsequently provide some evidence that this clear demarcation is not simply an incidental reflection of body size, but likely causally related. This evidence comes in the flavour of requiring Ps to imagine themselves as small as a cat or as large as an elephant and showing a predicted shift in the affordance boundary. The manuscript further demonstrates that ChatGPT (theoretically interesting because it's trained on language alone without sensorimotor information; trained now on words rather than images) showed a similar boundary.

The authors also conducted a small MRI study task where Ps decide whether a probe action was affordable (graspable?) and created a congruency factor according to the answer (yes/no). There was an effect of congruency in posterior fusiform and superior parietal lobule for objects within body size range, but not outside. No effects in LOC or M1.

The major strength of this manuscript in my opinion is the methodological novelty. I felt the correlation matrices were a clever method for demonstrating these demarcations, the imagination manipulation was also exciting, and the ChatGPT analysis provided excellent food for thought. These findings are important for our understanding of the interactions between action and perception, and hence for researchers from a range of domains of cognitive neuroscience.

The major element that limits conclusions is that an MRI study with 12 P in this context can really only provide pilot data. Certainly the effects are not strong enough for 12 P to generate much confidence. The others of my concerns have been addressed in the revision.

<https://doi.org/10.7554/eLife.90583.2.sa2>

Reviewer #2 (Public Review):

Summary

In this work, the authors seek to test a version of an old idea, which is that our perception of the world and our understanding of the objects in it are deeply influenced by the nature of our bodies and the kinds of behaviours and actions that those objects afford. The studies presented here muster three kinds of evidence for a discontinuity in the encoding of objects, with a mental "border" between objects roughly of human body scale or smaller, which tend to relate to similar kinds of actions that are yet distinct from the kinds of actions implied by human-or-larger scale objects. This is demonstrated through observers' judgments of the kinds of actions different objects afford; through similar questioning of AI large-language models (LLMs); and through a neuroimaging study examining how brain regions implicated in object understanding make distinctions between kinds of objects at human and larger-than-human scales.

Strengths

The authors address questions of longstanding interest in the cognitive neurosciences -- namely how we encode and interact with the many diverse kinds of objects we see and use in daily life. A key strength of the work lies in the application of multiple approaches. Examining the correlations among kinds of objects, with respect to their suitability for different action kinds, is novel, as are the complementary tests of judgments made by LLMs. The authors include a clever manipulation in which participants are asked to judge action-object pairs, having first adopted the imagined size of either a cat or an elephant, showing that the discontinuity in similarity judgments effectively moved to a new boundary closer to the imagined scale than the veridical human scale. The dynamic nature of the discontinuity hints that action affordances may be computed dynamically, "on the fly", during actual action behaviours with objects in the real world.

Weaknesses

A limitation of the tests of LLMs may be that it is not always known what kinds of training material was used to build these models, leading to a possible "black box" problem. Further, presuming that those models are largely trained on previous human-written material, it may not necessarily be theoretically telling that the "judgments" of these models about action-object pairs shows human-like discontinuities. Indeed, verbal descriptions of actions are very likely to mainly refer to typical human behaviour, and so the finding that these models demonstrate an affordance discontinuity may simply reflect those statistics, rather than providing independent evidence for affordance boundaries.

The relatively small sample size of the brain imaging experiment, and some design features (such as the task participants performed, and the relatively narrow range of objects tested) provide some limits on the extent to which it can be taken as support for the authors' claims.

<https://doi.org/10.7554/eLife.90583.2.sa1>

Reviewer #3 (Public Review):

Summary:

Feng et al. test the hypothesis that human body size constrains the perception of object affordances, whereby only objects that are smaller than the body size will be perceived as useful and manipulable parts of the environment, whereas larger objects will be perceived as "less interesting components."

To test this idea, the study employs a multi-method approach consisting of three parts:

In the first part, human observers classify a set of 24 objects that vary systematically in size (e.g., ball, piano, airplane) based on 14 different affordances (e.g., sit, throw, grasp). Based on the average agreement of ratings across participants, the authors compute the similarity of affordance profiles between all object pairs. They report evidence for two homogenous object clusters that are separated based on their size with the boundary between clusters roughly coinciding with the average human body size. In follow-up experiments, the authors show that this boundary is larger/smaller in separate groups of participants who are instructed to imagine themselves as an elephant/cat.

In the second part, the authors ask different large language models (LLMs) to provide ratings for the same set of objects and affordances and conduct equivalent analyses on the obtained data. Some, but not all, of the models produce patterns of ratings that appear to show similar boundary effects, though less pronounced and at a different boundary size than in humans.

In the third part, the authors conduct an fMRI experiment. Human observers are presented with four different objects of different sizes and asked if these objects afford a small set of specific actions. Affordances are either congruent or incongruent with objects. Contrasting brain activity on incongruent trials against brain activity on congruent trials yields significant effects in regions within the ventral and dorsal visual stream, but only for small objects and not for large objects.

The authors interpret their findings as support for their hypothesis that human body size constrains object perception. They further conclude that this effect is cognitively penetrable, and only partly relies on sensorimotor interaction with the environment (and partly on linguistic abilities).

Strengths:

The authors examine an interesting and relevant question and articulate a plausible (though somewhat underspecified) hypothesis that certainly seems worth testing. Providing more detailed insights into how object affordances shape perception would be highly desirable. Their method of analyzing similarity ratings between sets of objects seems useful and the multi-method approach is original and interesting.

Weaknesses:

The study presents several shortcomings that clearly weaken the link between the obtained evidence and the drawn conclusions. Below I outline my concerns in no particular order:

(1) It is not entirely clear to me what the authors are proposing and to what extent the conducted work actually speaks to this. For example, in the introduction, the authors write

that they seek to test if body size serves not merely as a reference for object manipulation but also "plays a pivotal role in shaping the representation of objects." This motivation seems rather vague motivation and it is not clear to me how it could be falsified.

Overall, the lack of theoretical precision makes it difficult to judge the appropriateness of the approaches and the persuasiveness of the obtained results. I would strongly suggest clarifying the theoretical rationale and explaining in more detail how the chosen experiments allow them to test falsifiable predictions.

(2) The authors used only a very small set of objects and affordances in their study and they do not describe in sufficient detail how these stimuli were selected. This renders the results rather exploratory and clearly limits their potential to discover general principles of human perception. Much larger sets of objects and affordances and explicit data-driven approaches for their selection would provide a more convincing approach and allow the authors to rule out that their results are just a consequence of the selected set of objects and actions.

(3) Relatedly, the authors could be more thorough in ruling out potential alternative explanations. Object size likely correlates with other variables that could shape human similarity judgments and the estimated boundary is quite broad (depending on the method, either between 80 and 150 cm or between 105 to 130 cm). More precise estimates of the boundary and more rigorous tests of alternative explanations would add a lot to strengthen the authors' interpretation.

(4) While I appreciate the manipulation of imagined body size, as a clever way to solidify the link between body size and affordance perception, I find it unfortunate that it is implemented in a between-subjects design, as this clearly leaves open the possibility of pre-existing differences between groups. I certainly disagree with the authors' statement that their findings suggest "a causal link between body size and affordance perception."

(5) The use of LLMs in the current study is not clearly motivated and I find it hard to understand what exactly the authors are trying to test through their inclusion. As it currently stands, I find it hard to discern how the presence of perceptual boundaries in LLMs could constitute evidence for affordance-based perception.

(6) Along the same lines, the fMRI study also provides little evidence to support the authors' claims. The use of congruency effects as a way of probing affordance perception is not well motivated. Importantly (and related to comment 2 above), the very small set of objects and affordances in this experiment heavily complicates any conclusions about object size being the crucial variable determining the occurrence of congruency effects.

Overall, I consider the main conclusions of the paper to be far beyond the reported data. Articulating a clearer theoretical framework with more specific hypotheses as well as conducting more principled analyses on more comprehensive data sets could help the authors obtain stronger tests of their ideas.

<https://doi.org/10.7554/eLife.90583.2.sa0>